



UNIVERSITA' DEGLI STUDI DI GENOVA

DIBRIS

DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY,
BIOENGINEERING, ROBOTICS AND SYSTEM ENGINEERING

COGNITIVE ARCHITECTURES FOR ROBOTICS

FINAL PROJECT

A-to-B Navigation with G1 EDU in Gazebo

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1.Introduction

This project develops a standalone A-to-B indoor navigation benchmark for the Unitree G1 EDU robot in Gazebo using ROS 2 Humble and Nav2. The robot is modeled with 29 degrees of freedom and uses a Livox MID-360 LiDAR and an Intel RealSense D435i depth camera.

The system performs navigation on a known map with simulator-based localization, global path planning, local obstacle avoidance, and recovery behaviors. Three local planner—DWB, MPPI, and RPP—are compared in corridor, doorway, clutter, dynamic blockage, and maze scenarios.

Performance is evaluated using success rate, completion time, path efficiency, collision events, recovery count, and final goal error. The main objective is to identify the most suitable planner configuration for reliable indoor navigation of the Unitree G1 platform.

2. Project Objective and Scope

The goal was not simply to demonstrate one successful navigation run. The project was designed as a controlled benchmark that could answer a practical question:

Which Nav2 local planner is most suitable for a humanoid-sized robot abstraction moving through structured indoor environments?

The implemented pipeline included a known map, simulator-provided baseline localization, global route generation, local obstacle-aware control, execution monitoring, recovery behavior, logging, and automatic metric generation. The Unitree G1 abstraction used a MID-360-style LiDAR and a D435i-style depth camera. The LiDAR point cloud was adapted to a 2D LaserScan stream for Nav2.

Implemented components

Component	Implementation
Simulation	Gazebo with the Unitree G1 EDU abstraction
Global planning	ThetaStar for all planner profiles
Local control	DWB, MPPI, and Regulated Pure Pursuit
Localization	Simulator baseline map-to-odom and odometry
Recovery	Nav2 behavior-tree recovery after progress failure
Evaluation	CSV telemetry, summary metrics, and plots

3. Cognitive and Software Architecture

The architecture is best described as hybrid reactive-deliberative. The main navigation chain is hierarchical: sensor information updates the navigation state, the goal is represented as a GO-TO task, the Robot Execution Manager coordinates global planning and local control, and execution is monitored against the expected result. If the robot reaches the goal, the task succeeds; if progress is lost, the system can trigger recovery.

The reactive safety path is separate from the slower deliberative loop but remains inside the Act layer. Immediate obstacle information can feed a Braitenberg-inspired reflex, whose safe command competes with the normal planner command through a subsumption coordinator. This preserves the three core course layers - Sense, Plan/Reasoning, and Act - while adding a fast sensory-motor safety behavior.

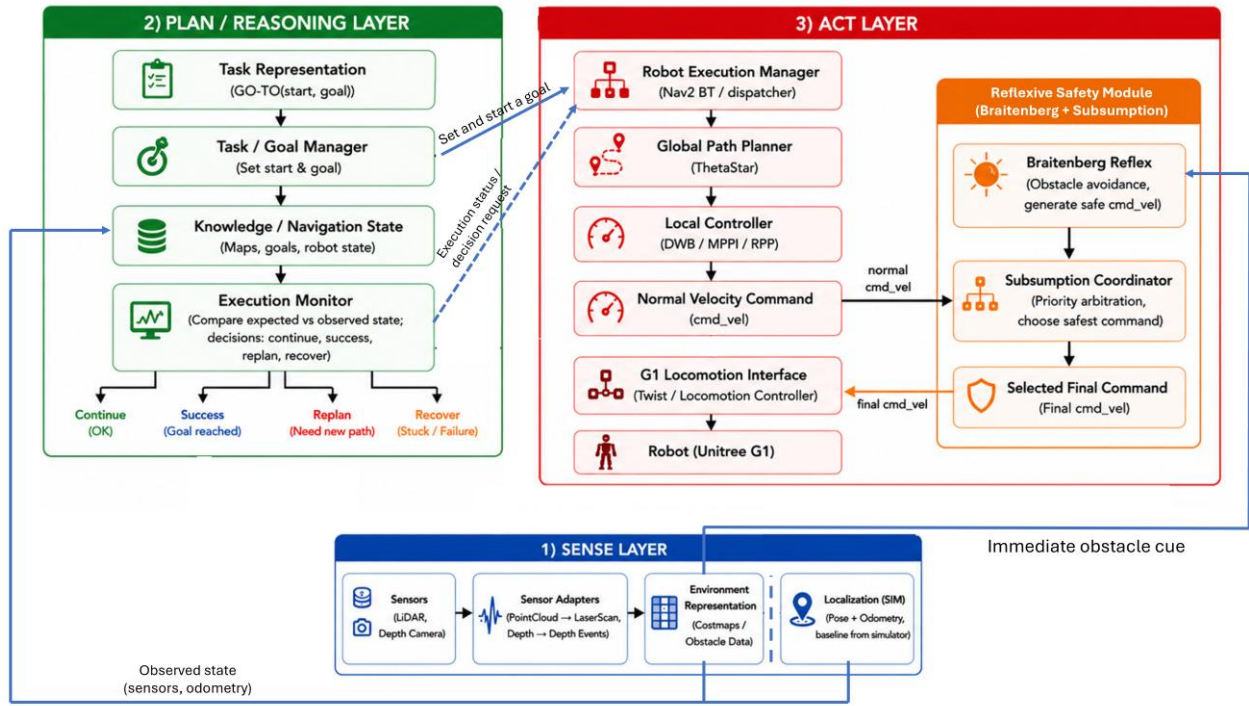


Figure 1. Hierarchical navigation with execution monitoring and fast bio-inspired safety.

3. Experimental Design

A fair comparison required the environment and global planning conditions to remain fixed. All three local planners used the same maps, start and goal poses, robot footprint, costmaps, velocity limits, goal tolerance, startup procedure, timeout, and ThetaStar global planner. Only the local planner profile changed.

The benchmark matrix contained five scenarios, three planner profiles, and three repetitions, for a total of 45 runs. The scenarios were selected to increase difficulty gradually from basic timing to dynamic rerouting and high-curvature navigation.

Scenario	Purpose	Main challenge
Corridor	Baseline timing	Straight, open navigation
Doorway	Narrow-passage test	Footprint and alignment
Clutter	Static obstacle adaptation	Local trajectory selection
House dynamic	Dynamic rerouting	Route blockage during execution
Maze	Complex route following	High curvature and constrained geometry

Metrics and evaluation priority

The report treats success rate and completion time as the primary criteria, because they directly answer whether the task was completed and how efficiently it was completed. Final goal error, recovery count, minimum clearance, path efficiency, and collision proxy events were used as supporting measures.

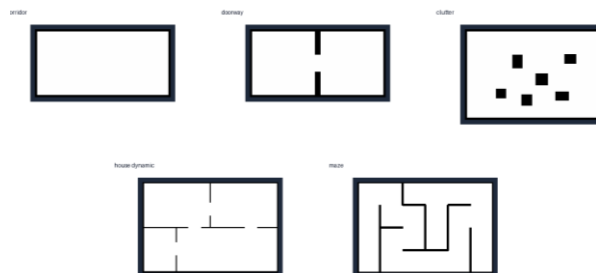


Figure 2. Five indoor tasks stress different navigation behaviors.

4. Primary Results: Success and Completion Time

The main result is clear. All planners completed the three easier scenarios. The separation appeared in house_dynamic and maze: MPPI and RPP completed every run, while DWB failed all repetitions in both scenarios. Across the full benchmark, this corresponds to 100% overall success for MPPI and RPP and 60% for DWB.

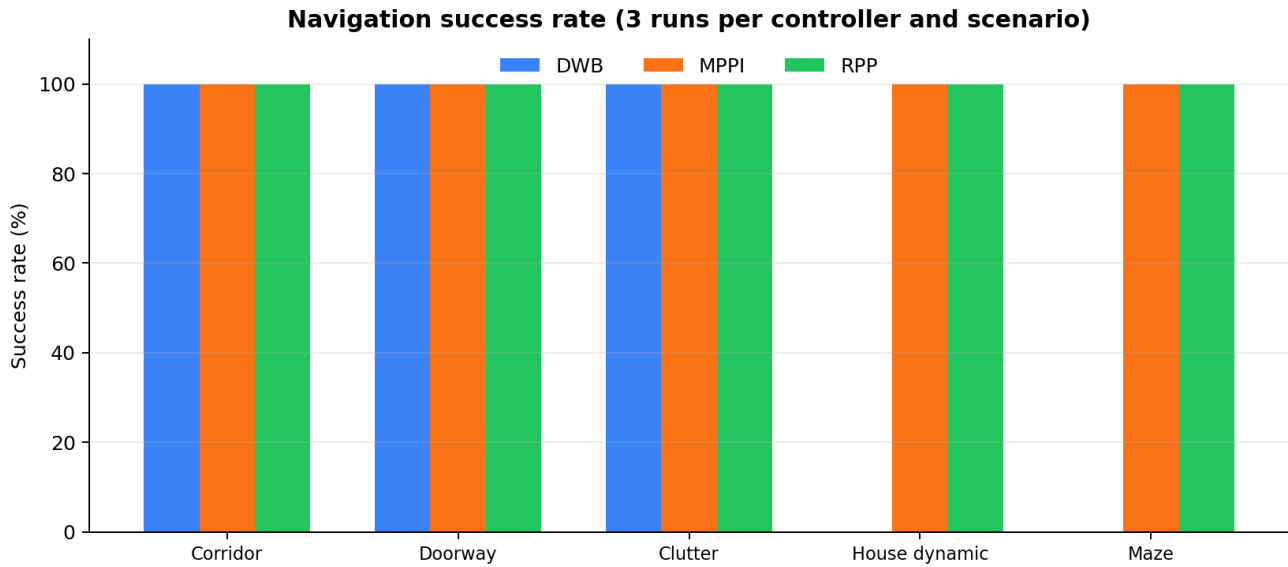


Figure 3. Success rate by planner and scenario. DWB failed consistently in the two hardest tasks.

Completion time reinforces the same conclusion. In the complex successful cases, MPPI was faster than RPP: approximately 69.1 s versus 76.9 s in house_dynamic, and 136.1 s versus 150.1 s in maze. DWB approached the run timeout in both failed scenarios, with mean durations of approximately 185.0 s and 196.1 s.

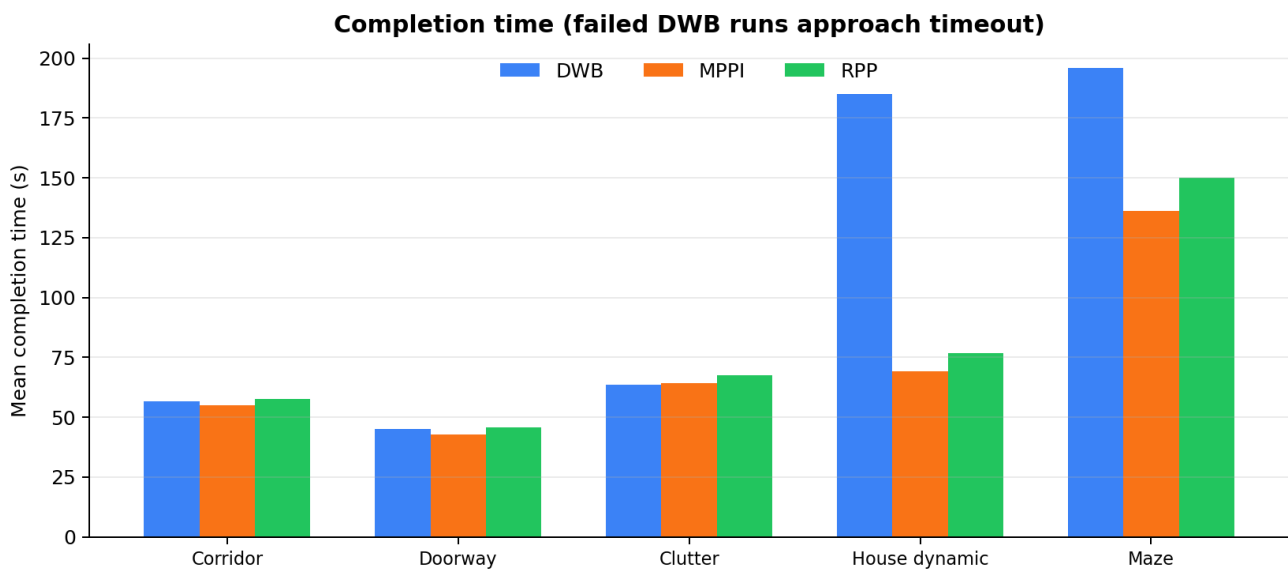


Figure 3. Mean completion time. The high DWB values in the hard scenarios correspond to failed runs approaching timeout.

Interpretation

MPPI did not only succeed more reliably; it also preserved competitive completion time in complex geometry. This supports the expected advantage of predictive, sampled control when local decisions become difficult.

5. Recovery, Accuracy, and Safety

DWB did not fail because of a recorded obstacle-contact event. Its main failure mode in the difficult scenarios was loss of progress in sharp turns and high-curvature sections. It triggered an average of two recovery actions in `house_dynamic` and three in `maze`. However, the available recovery sequence did not move the robot into a sufficiently different state to escape the local minimum and resume navigation. This is a valid benchmark finding: recovery was triggered correctly, but it was not effective for the specific failure condition encountered by DWB.

MPPI achieved the smallest final goal error in every scenario. RPP also completed every task, although it generally stopped with a slightly larger residual error. No collision-proxy events were recorded in the final aggregate results. However, this metric is based on obstacle proximity and should not be interpreted as physical-contact ground truth.

Planner	Overall success	Hard-case recovery	Braitenberg reflex activation	Goal accuracy	Assessment
DWB	60%	2 in <code>house_dynamic</code> ; 3 in <code>maze</code>	1 activation per run in <code>house_dynamic</code>	Large error on failed runs	Useful baseline; weak in hard cases
MPPI	100%	No Nav2 recovery required	1 activation per run in <code>house_dynamic</code>	Best in all scenarios	Best overall profile
RPP	100%	No Nav2 recovery required	1 activation per run in <code>house_dynamic</code>	Reliable, slightly less precise	Robust but conservative

Recovery interpretation

Recovery should not be confused with normal obstacle avoidance or reactive safety. In the `house_dynamic` scenario, a door-like obstacle suddenly appeared and blocked the current route. The Braitenberg-inspired reflex responded to the immediate obstacle cue and temporarily modified the velocity command, redirecting the robot away from the obstacle and preventing an unsafe approach.

The updated obstacle information was then incorporated into the navigation system. Global replanning generated an alternative route, and the local planners followed it. DWB, MPPI, and RPP were all able to respond correctly to the dynamic blockage itself. Therefore, the recovery events recorded for DWB were not caused directly by the appearance of the door. They occurred later, when DWB encountered sharp turns or high-curvature sections and could no longer maintain sufficient progress.

In these cases, Nav2 correctly detected the loss of progress and activated recovery. However, the available recovery behaviors did not produce enough positional change to move the robot out of the local minimum. As a result, DWB could return to nearly the same configuration and fail again.

Therefore, the three mechanisms had distinct roles: the Braitenberg reflex provided immediate local safety, the global planner generated an alternative route around the blockage, and recovery was activated only when normal navigation later failed to make progress.

6. Discussion

The results support a planner ranking for this specific simulation configuration, not a universal claim. MPPI was the strongest overall choice because it combined complete task success, competitive time, and the smallest final error. RPP was a strong practical alternative: it was safe and reliable, but its regulated behavior was slower and slightly less precise. DWB remained valuable as a transparent baseline, but its tuning and recovery interaction were not sufficient for the dynamic and maze tasks.

From a COGAR perspective, the project is more than a planner comparison. It demonstrates how symbolic task intent, metric planning, continuous control, execution monitoring, recovery, and fast reactive behavior can be integrated into a reproducible ROS 2 system. The Braitenberg-inspired reflex was exercised in the `'house_dynamic'` scenario, where it provided an immediate sensory-motor safety response to the suddenly

appearing obstacle. Its role was to modify the short-term velocity command, while global replanning and local planner remained responsible for generating and following the alternative route.

7. Limitations and Future Work

- The evaluation is simulation-only and uses simulator-provided baseline localization.
- Each planner-scenario pair was repeated three times; more repetitions would strengthen statistical confidence.
- The locomotion model was intentionally simplified because the main objective of the project was navigation planning, obstacle avoidance, and controller benchmarking rather than full-body humanoid motion control. Realistic locomotion for a 29-DOF humanoid would require whole-body dynamics, balance control, gait generation, and significantly greater computational resources.
- Future work should include real G1 trials, moving-human scenarios, parameter sensitivity analysis, and failure-specific recovery behaviors.

8. Conclusion

This work produced a complete and repeatable A-to-B navigation benchmark for the Unitree G1 EDU abstraction in Gazebo. By fixing ThetaStar global planner and varying only the local planner, the experiments isolated meaningful differences between DWB, MPPI, and RPP. The primary criteria - success and completion time - identify MPPI as the strongest overall profile in the tested scenarios. RPP was equally reliable but more conservative, while DWB failed in the two hardest tasks despite triggering recovery.

The final contribution is therefore both engineering and cognitive: a reproducible ROS 2 benchmark, and a clear mapping from COGAR concepts to a working hybrid architecture with hierarchical planning, execution monitoring, recovery, and reactive safety.

Final takeaway

MPPI is the recommended planner for the final G1 benchmark configuration. RPP is a robust alternative. DWB should remain as the baseline and as evidence of where the current planner/recovery combination loses progress.

References

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